

ESP32-S3 N16R8 — Verified Hardware Profile

Purpose: Permanent reference for this specific third-party ESP32-S3 development board. The profile records measurements and observations already obtained so the hardware does not need to be re-audited before future ESP-IDF/FreeRTOS work.

1. Executive Summary

Item	Verified result
MCU	ESP32-S3, QFN56, silicon revision v0.2
Module marking	ESP32-S3-WROOM / N16R8 (visual observation from board)
Flash	16 MB physically detected
PSRAM	8 MB physically detected and stress-tested
CH343 UART	WCH USB-UART bridge, COM7
Native USB	ESP32-S3 USB VID 303A / PID 1001; USB Serial + USB JTAG/serial debug unit, COM8
Crystal	40 MHz
ESP-IDF	v6.0.2
Project target	esp32s3

2. MCU / Silicon

ESP32-S3: Direct esptool identification reported ESP32-S3 (QFN56), revision v0.2. The chip identification also reported Wi-Fi, Bluetooth 5 LE, dual core + LP core, and a 240 MHz capable CPU. Crystal frequency was reported as 40 MHz. These are direct device-identification results, not inferred from the PCB label.

MAC address observed: CC:8D:A2:EC:FF:BC.

Important: The MAC address is a hardware identity value and should not be published unnecessarily in public repositories.

3. Flash Memory

Property	Verified value	Evidence/status
Capacity	16 MB	Direct flash-id detection; later boot log also reports 16 MB
Manufacturer ID	0x85 (decimal 85)	Direct flash-id result
Device ID	0x2018	Direct flash-id result
Flash voltage	3.3 V	Direct flash-id result
Flash interface	Quad / DIO configuration	eFuse/boot configuration and application boot log
Configured mode	DIO	sdkconfig/build configuration
Configured frequency	80 MHz	sdkconfig/build configuration

Note: the exact commercial flash part-number was not independently mapped from the JEDEC/device IDs in the collected audit. For practical development, capacity, interface, and operating configuration are verified.

4. PSRAM

Property	Verified value
Capacity	8 MB (8,388,608 bytes)
Density	64 Mbit

Vendor ID	0x0D, reported as AP
Generation	Generation 3
Die status	Good-die / Pass
Voltage	3 V reported by PSRAM identification
Mode	Octal
Speed	40 MHz
Latency	Fixed; read latency reported as 10 cycles
Burst	Hybrid Wrap, 32-byte burst
Drive strength	1/1
ESP-IDF heap integration	8,192 KB PSRAM pool
PSRAM memory test	PASS
Custom stress test	7 × 1 MB blocks allocated; 0xAA, 0x55 and address-dependent read/write tests passed

The custom stress test triggered the Task Watchdog during a long CPU-bound verification loop, but the address-dependent memory verification subsequently passed. This was a task-yielding/watchdog issue, not a PSRAM integrity failure.

5. ESP-IDF Project Configuration

Configuration	Verified value
ESP-IDF	v6.0.2
Target	esp32s3
Architecture	Xtensa
Flash mode	DIO
Flash size	16 MB
Flash frequency	80 MHz
PSRAM enabled	CONFIG_SPIRAM=y
PSRAM mode	Octal
PSRAM speed	40 MHz
PSRAM boot initialization	Enabled
PSRAM heap allocation	Enabled
PSRAM memtest	Enabled
PSRAM clock IO	GPIO30
PSRAM CS IO	GPIO26
PSRAM XIP	Not enabled

6. USB Interfaces

Interface	Verified result
USB-UART bridge	WCH CH343
CH343 VID/PID	VID 1A86 / PID 55D3
CH343 manufacturer	wch.cn
CH343 port	COM7
Native ESP32-S3 USB VID/PID	VID 303A / PID 1001
Native USB serial	USB Serial Device (COM8)
Native USB debug	USB JTAG/serial debug unit

The second USB-C connection was independently observed by Windows as VID 303A/PID 1001 and exposed both a USB Serial Device (COM8) and a USB JTAG/serial debug unit. This confirms that the board exposes the ESP32-S3's native USB interface in the tested firmware state.

7. Security / eFuse State

Previously collected esptool security information reported:

Property	Observed value
Secure Boot	Disabled
Flash Encryption	Disabled
SPI_BOOT_CRYPT_CNT	0x0
Security flags	0x00000000
Chip ID from security information	9
API version	0

8. Serial / Development Connections

Primary UART development path: USB → CH343 → COM7 → ESP32-S3 UART console.

Native USB path: USB → ESP32-S3 native USB → COM8 + JTAG/serial debug unit when the current firmware exposes it.

Known Windows device: USB-Enhanced-SERIAL CH343 (COM7), status OK, WCH manufacturer.

9. Board Identity / Authenticity Assessment

The PCB itself is a third-party development board rather than an official Espressif development board. However, the important ESP32-S3 N16R8 hardware claims were independently corroborated: the MCU identifies as ESP32-S3, physical flash reports 16 MB, PSRAM identifies as an 8 MB AP octal device, the ESP-IDF PSRAM test passes, and a custom 7 MB read/write stress test passes. The module marking N16R8 is therefore consistent with measured hardware behavior.

Practical verdict: The board is sufficiently verified for embedded development, ESP-IDF, FreeRTOS, Wi-Fi/BLE, native USB, and memory-intensive experiments. There is no practical reason to repeat the hardware audit before starting RTOS work.

10. Important Caveats / Not Independently Established

- Exact commercial flash part number was not mapped beyond manufacturer/device IDs.
- Exact PCB manufacturer/model was not established from the collected software audit.
- Complete board-specific GPIO schematic/pinout was not independently traced.

- Regulator part number and detailed power-rail measurements were not independently measured.
- The native USB interface was confirmed as enumerating, but every possible native USB function was not exhaustively tested.
- The custom PSRAM stress loop can trigger the FreeRTOS Task Watchdog if it runs for too long without yielding; this is a software scheduling issue.

11. Useful Development Constants

Project: D:\Project\ESP32\hello_world

ESP-IDF path: C:\esp\v6.0.2\esp-idf

ESP-IDF Python: C:\Espressif\tools\python\v6.0.2\venv\Scripts\python.exe

UART port: COM7

Native USB serial port observed: COM8

Target: esp32s3

12. Recommended Starting Point for RTOS

Hardware audit can now be considered closed. Keep this PDF as the board reference. For future work, use the verified ESP32-S3 target and do not change Flash/PSRAM settings unless a project specifically requires it. The next learning path can proceed directly into FreeRTOS concepts: tasks, scheduler, priorities, blocking/delay, queues, semaphores, mutexes, event groups, software timers, task notifications, ISR/task interaction, and watchdog behavior.

Source basis: user-provided ESP-IDF/esptool/Windows PnP logs and the completed PSRAM verification test from this conversation. Values marked as observed/verified are limited to those logs; unmeasured PCB details are intentionally left as caveats.